AI CODE ARCHITECT

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INTRODUCTION

AI Code Architect is an intelligent system that generates, analyzes, and optimizes code using Artificial Intelligence.

It helps developers by automating coding tasks, reducing errors, and improving productivity.

OBJECTIVES

To develop an AI-based code generation system

To automate programming tasks

To reduce development time

To improve code quality and efficiency

To assist beginners in coding

STAKEHOLDERS

Software Developers

Students & Beginners

IT Companies

Researchers

Educators

TECHNOLOGIES USED

Programming Language: Python

Frontend: HTML, CSS, JavaScript

Backend: Flask / Node.js

Concepts:

Artificial Intelligence

Natural Language Processing (NLP)

AI TOOLS USED

ChatGPT – Code generation

Google Gemini – Alternative AI support

GitHub Copilot – Code suggestions

PROMPTS FOR AI CODE ARCHITECT

- "Generate Python code for AI Code Architect system"
- "Create a program that generates code based on user input"
- Explain the generated code step by step
- Identify and fix errors in the code

AI CODE ARCHITECT – WORKFLOW FLOWCHART

Start

↓

User Input (Prompt)

↓

Send to AI Tool (ChatGPT / Gemini)

↓

AI Processing (NLP Understanding)

↓

Code Generation

↓

Display Generated Code

↓

User Verification / Debugging

↓

FRONT-END VIEW OF APPLICATION

Basic Components:

Input box (where user types prompt)

Submit / Generate button

Output area (where code is shown)

File Edit Selection View Go Run ···

Search

EXPLORER

OPEN EDITORS 1 unsaved

Untitled-1

AI.py C:\Users\koush

NO FOLDER OPENED

You have not yet opened a folder.

Open Folder

Opening a folder will close all currently open editors. To keep them open, add a folder instead.

You can clone a repository locally.

Clone Repository

To learn more about how to use Git and source control in VS Code read our docs.

OUTLINE

Untitled-1 | AI.py

C: > Users > koush > AI.py

```
#generate a python code for AI code Architect
```

CHAT

SESSIONS

Tip: Try the Plan agent to research and plan before implementing changes.

AI.py

Describe what to build

Auto

Local Default Approvals

0 0 Python Debugger: Python File Ln 1, Col 46 Spaces: 4 UTF-8 CRLF Python Select Python Interpreter

32°C Sunny Search ENG IN 13:09 05-04-2026

REFERENCES

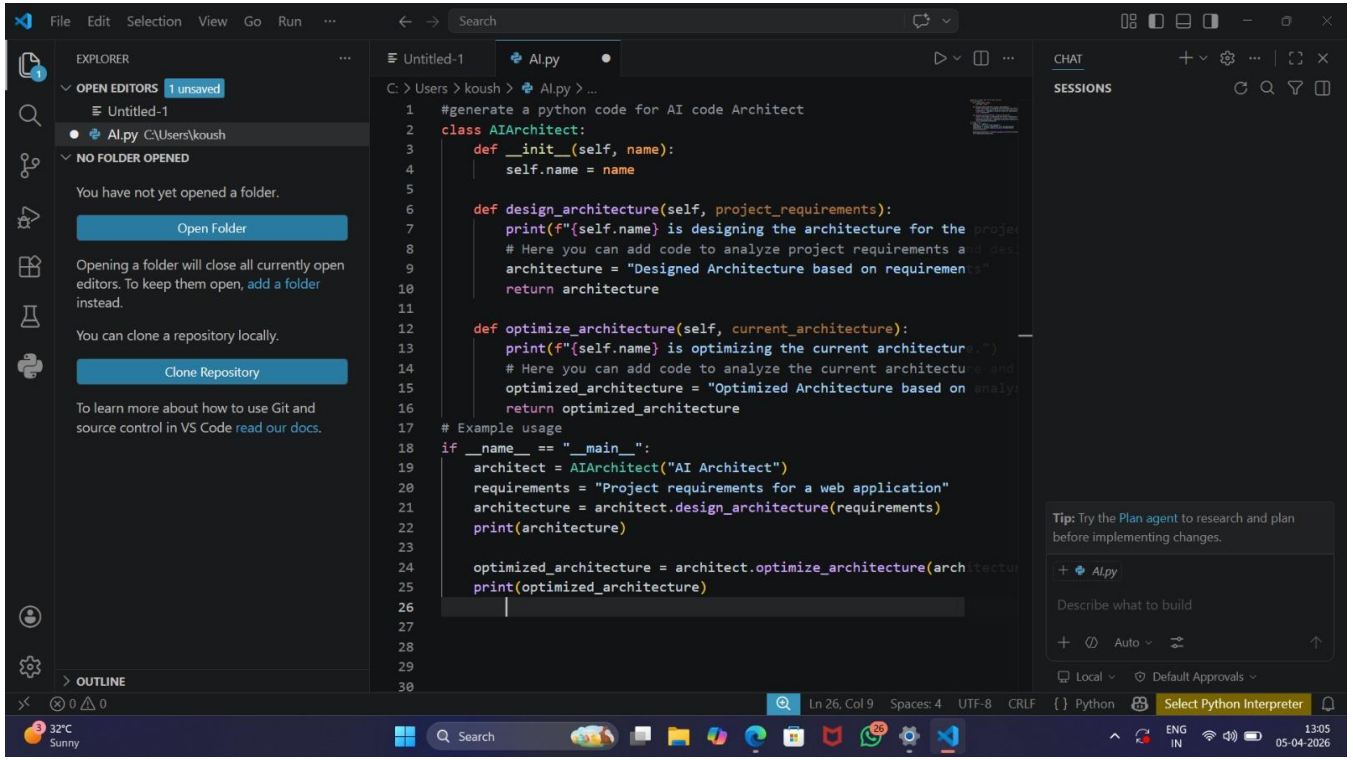
ChatGPT

GitHub

Google

Python Documentation

CODE


File Edit Selection View Go Run
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EXPLORER
OPEN EDITORS 1 unsaved
Untitled-1
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OUTLINE
Untitled-1
AI.py
C: > Users > koush > AI.py > ...
#generate a python code for AI code Architect
class AIArchitect:
def __init__(self, name):
self.name = name
def design_architecture(self, project_requirements):
print(f"{self.name} is designing the architecture for the
# Here you can add code to analyze project requirements a
architecture = "Designed Architecture based on requiremen
return architecture
def optimize_architecture(self, current_architecture):
print(f"{self.name} is optimizing the current architectur
# Here you can add code to analyze the current architectu
optimized_architecture = "Optimized Architecture based on
return optimized_architecture
# Example usage
if __name__ == "__main__":
architect = AIArchitect("AI Architect")
requirements = "Project requirements for a web application"
architecture = architect.design_architecture(requirements)
print(architecture)
optimized_architecture = architect.optimize_architecture(arch
print(optimized_architecture)
CHAT
SESSIONS
Tip: Try the Plan agent to research and plan before implementing changes.
AI.py
Describe what to build
Auto
Local
Default Approvals
Ln 26, Col 9
Spaces: 4
UTF-8
CRLF
Python
Select Python Interpreter
32°C Sunny
Search
ENG IN
13:05 05-04-2026

CONCLUSION

The project successfully demonstrates AI-based code generation

It reduces manual effort

Useful for beginners and developers

Can be enhanced with real AI APIs in future